

TORONTO BUTTONVILLE, ON, Canada

WMO: 716390

Lat: 43.8660N

Lon: 79.3681W

Elev: 198

StdP: 98.97

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-16.9	-26.1	0.4	-19.0	-22.8	0.5	-16.3	10.8	-6.0	9.8	-6.0	2.9	340	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.6	22.3	29.8	21.2	28.1	20.5	23.5	29.3	22.5	27.8	21.6	26.4	4.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	16.8	26.2	20.8	15.8	25.3	19.9	14.9	24.6	71.2	29.5	67.3	28.0	64.0	26.3	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.3	7.5	-23.7	34.4	3.9	1.9	-26.5	35.8	-28.8	36.9	-31.0	37.9	-33.8	39.3		
			-24.1	25.3	3.8	1.3	-26.8	26.3	-29.1	27.0	-31.2	27.7	-34.0	28.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-5.8	-5.0	-0.1	6.6	13.4	18.9	21.6	20.7	16.7	9.8	3.5	-2.0
	DBStd	10.74	6.52	5.70	5.74	4.65	4.57	3.78	3.13	3.01	3.94	4.27	4.57	5.18
	HDD10.0	1997	491	419	317	122	17	0	0	0	1	57	199	373
	HDD18.3	3967	749	653	572	353	167	37	6	12	77	264	446	631
	CDD10.0	1358	0	0	3	19	121	268	359	330	202	52	4	0
	CDD18.3	287	0	0	0	0	13	55	107	84	27	1	0	0
	CDH23.3	2426	0	0	1	8	143	491	927	619	223	13	0	0
	CDH26.7	705	0	0	0	1	35	156	295	161	56	2	0	0
Wind	WSAvg	3.4	3.8	3.8	3.8	3.9	3.4	3.1	3.0	2.7	2.8	3.1	3.4	3.6
Precipitation	PrecAvg	804	61	49	55	72	75	75	83	73	65	66	70	60
	PrecMax	979	134	96	114	128	141	187	148	147	155	134	134	113
	PrecMin	582	27	18	22	24	22	11	14	17	12	20	29	22
	PrecStd	111	24	23	25	27	27	43	32	28	32	32	28	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	11.2	20.5	24.4	30.0	33.2	34.0	33.1	31.5	24.9	17.5	12.4
		MCWB	9.6	7.8	13.9	15.9	20.1	23.0	23.6	22.7	22.1	18.5	13.1	10.1
	2%	DB	7.0	7.0	14.5	20.4	27.2	30.6	31.9	30.2	28.0	21.8	14.6	9.1
		MCWB	5.4	4.6	9.7	13.3	18.6	21.4	22.5	21.7	20.7	16.6	11.8	7.4
	5%	DB	4.5	4.5	10.8	17.2	24.3	28.3	29.8	28.3	25.9	19.1	12.6	6.7
		MCWB	3.2	2.6	7.0	10.9	16.9	20.4	21.4	20.8	19.4	15.0	9.9	5.0
	10%	DB	2.6	2.6	7.6	14.4	21.7	26.1	28.0	26.8	23.7	16.8	10.4	4.6
		MCWB	1.3	0.9	4.3	9.0	15.2	19.0	20.4	19.9	18.2	13.5	8.1	3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	8.1	15.3	17.2	22.1	24.2	25.3	24.4	23.5	20.2	14.4	10.6
		MCDB	10.9	10.9	19.5	22.6	28.1	31.6	31.5	29.3	28.9	23.4	15.9	11.9
	2%	WB	5.7	5.0	10.7	14.6	19.8	22.5	23.7	23.0	21.9	17.4	12.3	7.7
		MCDB	6.7	6.6	13.6	18.9	25.6	28.8	29.5	28.4	26.5	20.8	14.4	8.9
	5%	WB	3.3	2.7	7.3	11.9	17.9	21.2	22.6	22.0	20.5	15.6	10.1	5.3
		MCDB	4.4	4.1	10.7	16.3	23.4	26.8	27.8	26.4	24.1	18.3	12.0	6.5
	10%	WB	1.4	1.1	4.8	9.6	16.0	20.0	21.6	21.1	19.2	14.1	8.3	3.2
		MCDB	2.4	2.4	7.3	13.4	20.4	24.4	26.4	25.4	22.8	16.4	10.2	4.5
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.2	8.7	9.9	10.7	10.4	10.5	10.2	10.4	8.9	7.4	6.5
		MCDBR	8.2	9.7	13.3	15.0	14.1	13.3	12.6	12.1	13.2	12.6	10.8	8.3
		MCWBR	7.8	8.1	9.6	9.8	7.8	6.8	5.9	5.6	6.9	8.2	8.5	7.4
	5% WB	MCDBR	8.2	9.3	12.7	13.3	12.6	11.9	10.9	10.2	11.0	11.0	9.9	8.1
		MCWBR	8.0	8.4	9.7	9.8	7.8	6.7	5.7	5.4	6.5	7.9	8.6	7.8
Clear-Sky Solar Irradiance	taub	0.296	0.312	0.328	0.373	0.404	0.430	0.425	0.413	0.380	0.359	0.316	0.295	
	taud	2.375	2.328	2.403	2.319	2.282	2.236	2.269	2.292	2.370	2.411	2.487	2.431	
	Ebn at Noon	830	877	912	890	869	844	843	841	844	814	804	797	
	Edh at Noon	77	97	103	122	130	137	131	124	106	89	70	67	
All-Sky Solar Radiation	RadAvg	1.40	2.18	3.47	4.44	5.38	5.85	5.89	5.15	4.11	2.45	1.52	1.10	
	RadStd	0.14	0.21	0.35	0.46	0.57	0.43	0.42	0.26	0.38	0.23	0.19	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page